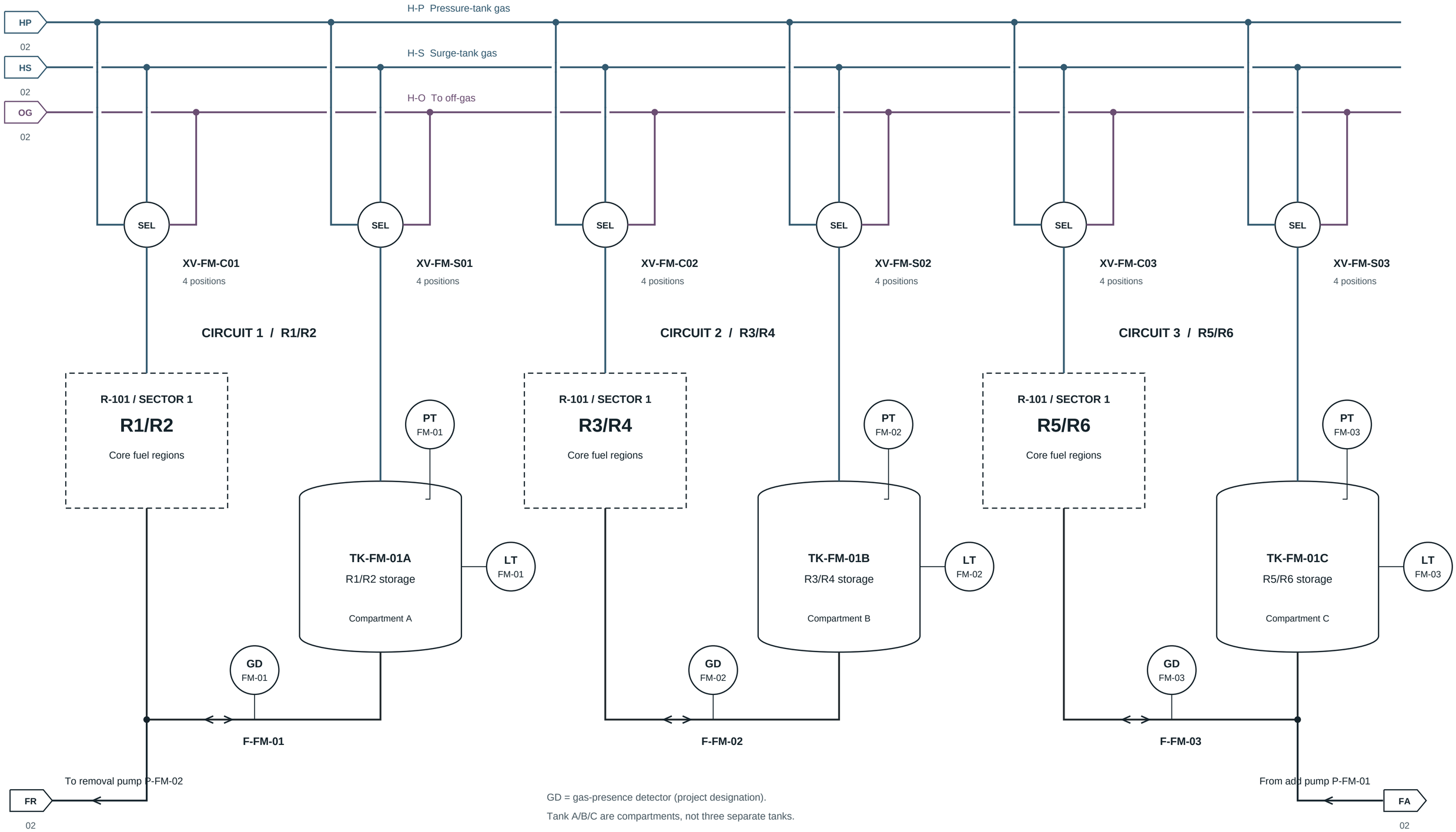
FAST SALT SYSTEM

FUEL TRANSFER AND PRESSURE VALVES

Sector 1 / TK-FM-01 (3D F6). Gas unit 3 is shared with sector 6 / TK-FM-06 (3D F5). Six-sector schedule: sheet 3.



GD = gas-presence detector (project designation).

Tank A/B/C are compartments, not three separate tanks.

Cooling-air connections for the common maintenance-tank enclosure are on sheet 2. No unshown liquid junction is implied within a core pair.

PRELIMINARY - FOR REVIEW

Owner-defined process arrangement. Sizes, ratings, fail positions and detailed controls TBD.

Functional schematic; not to scale. Solid-dot crossings connect; other crossings do not.

WISTR / FUEL MAINTENANCE

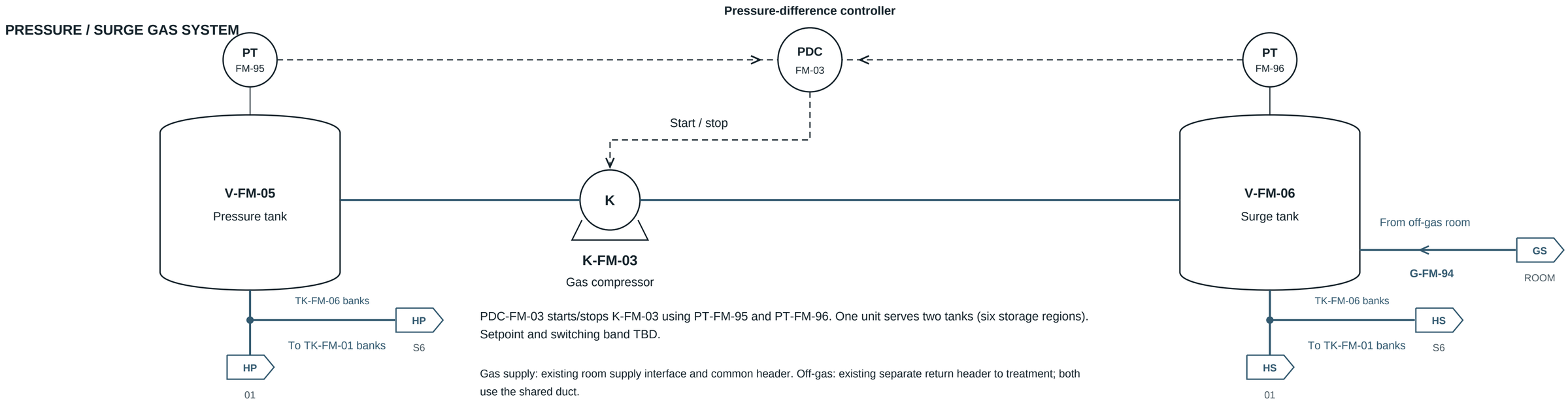
FS-FM-001 REV B3-draft 2026-10-15

FAST SALT SYSTEM

GAS SERVICES AND METERED FUEL CONNECTIONS

Connection markers HP / HS / OG / FA / FR continue on sheet 1. Gas unit 3 serves TK-FM-06 / TK-FM-01 (3D F5/F6); three shared units total.

PRESSURE / SURGE GAS SYSTEM

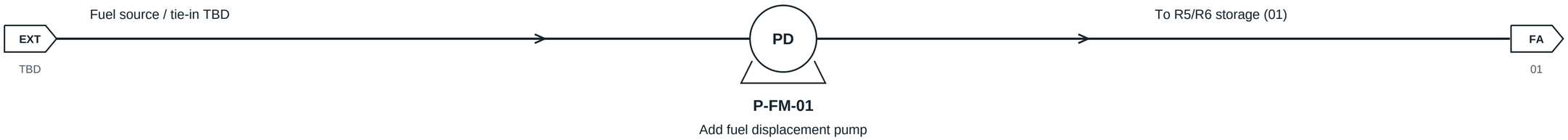


PDC-FM-03 starts/stops K-FM-03 using PT-FM-95 and PT-FM-96. One unit serves two tanks (six storage regions). Setpoint and switching band TBD.

Gas supply: existing room supply interface and common header. Off-gas: existing separate return header to treatment; both use the shared duct.

CORE / STORAGE VALVE VENT CONNECTION

METERED FUEL ADDITION AND REMOVAL



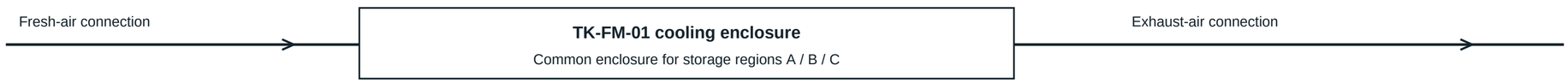
P-FM-01

Add fuel displacement pump

P-FM-02

Remove fuel displacement pump

COOLING AIR



Air connections retained from reference; detailed dampers, cooling internals and external air-header tags remain TBD.

PRELIMINARY - FOR REVIEW

Owner-defined process arrangement. Sizes, ratings, fail positions and detailed controls TBD.

Functional schematic; not to scale. Solid-dot crossings connect; other crossings do not.

WISTR / FUEL MAINTENANCE

FS-FM-002 REV B3-draft 2026-10-15

FAST SALT SYSTEM

VALVE POSITIONS, OPERATING STATES AND TAG SCHEDULE

Owner-defined states; compressor pressure-difference start/stop control on sheet 2. Setpoints and fail positions TBD.

SELECTOR POSITION SCHEDULE

Position	Core selectors XV-FM-Cxx	Storage selectors XV-FM-Sxx	Connection intent
1 - Add pressure	Available	Available	Common gas port C to pressure header H-P
2 - Maintain	Available	Available	Maintain position; internal port isolation details TBD
3 - Reduce pressure	Available	Available	Common gas port C to surge header H-S
4 - Vent to off-gas	Available	Operator selection only	Selected common gas port C to off-gas header H-O

OWNER-DEFINED OPERATING STATES

State / scope	Core pressure valves	Storage pressure valves	Pump / completion description
Remove fuel / all pairs	Add pressure	Reduce pressure	Gas in a bottom line: corresponding two valves return to Maintain. All valves in Maintain when all tanks are empty.
Fill fuel / all pairs	Reduce pressure	Add pressure	Gas in a maintenance-tank bottom line indicates empty: corresponding two valves return to Maintain. All valves in Maintain at completion.
Small addition / R5/R6	Reduce pressure	Add pressure	P-FM-01 meters fuel into R5/R6 storage. Use Fill fuel positions until gas is detected, then both R5/R6 valves return to Maintain. Intended addition to R6.
Measured removal / R1/R2	TBD	TBD	P-FM-02 removes a measured quantity from the R1/R2 core fuel line. Quantity, valve positions and further sequence are not specified.
Off-gas vent	Position 4, if selected	Position 4: operator only	Storage vent is used only when selected by the operator; excluded from automatic transfer sequences. Core vent selection conditions remain unspecified.

THREE SHARED GAS UNITS / EXISTING 3D ASSOCIATIONS

Unit / server	3D tanks	P&ID tanks	Sectors
1 / P1	F1 / F2	TK-FM-02 / TK-FM-03	2 / 3
2 / P2	F3 / F4	TK-FM-04 / TK-FM-05	4 / 5
3 / P3	F5 / F6	TK-FM-06 / TK-FM-01	6 / 1

Each tank: A=R1/R2, B=R3/R4, C=R5/R6. Circuit index = 3 x (sector - 1) + pair. Existing provisional sector/tank mapping retained (D-035).

R6-only distribution and metering-pump sharing remain TBD. Unit internals are detailed here; 3D servers remain exterior envelopes.

Notation reference: ISA5.1 public guidance; project-specific selector and GD conventions are defined here. No full-standard compliance claim.

SYMBOLS AND OPEN DETAILS

LT / PT: level / pressure sensing functions. GD: project-defined gas-presence detector. Circular multiport valve symbol: selector; actual actuation and internal trim TBD. PD pump: positive displacement.

Solid heavy lines: process piping. Solid thin lines: instrument process connections. Dashed lines: functional measurement/command signals. Dashed boxes: core boundaries. PDC: pressure-difference controller. Dot = junction; other crossings do not connect.

PRELIMINARY - FOR REVIEW

Owner-defined process arrangement. Sizes, ratings, fail positions and detailed controls TBD.

Functional schematic; not to scale. Solid-dot crossings connect; other crossings do not.

WISTR / FUEL MAINTENANCE

FS-FM-003 REV B3-draft 2026-10-15